Muhammad Zaid

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Summary

Computer Science student experienced in building software projects in Python and C++, including web publishing automation, game development, and multithreaded graphics programming. Strong foundation in software processes, architecture, design patterns, documentation, algorithms, data structures, version control and mathematics.

Skills

- **Programming Languages** C, C++, Python, JavaScript
- **Tools & Frameworks** Git, LaTeX, SFML, Pygame, SQLite
- **Concepts** Object Orientation, Design Patterns (Observer, Strategy), Version Control, CI/CD, Algorithms & Data structures, Unit Testing
- **Soft Skills** Abstract and critical thinking, analytical problem-solving, technical writing

Projects

Notes Publisher https://chuzawick420.github.io/notes_publisher/

- Architected an automated personal knowledge-base pipeline that converts Markdown notes written in Obsidian into a published website using MkDocs, Python, and LaTeX.
- Configured a GitHub Actions workflow to automatically build and deploy the site to GitHub Pages on every push, eliminating manual deployment steps.
- Designed an automated backup workflow using Google Drive, GitHub, and GitLab to provide redundant storage and reduce risk of data loss from hardware or power failures.

Memory Maze — Final Year Project https://github.com/ChuzaWick420/FYP_AI_Based_Puzzle_Game

- Designed and built a puzzle game engine in Python using a layered architecture, applying the Observer and Strategy design patterns to decouple game logic from the UI (made with Pygame) and persistent data.
- Built a JSON and SQLite-backed persistence layer to store game state and user data.
- Implemented time-slicing for input handling and event handling to keep the UI responsive during gameplay.

Ray Tracing https://github.com/ChuzaWick420/rayTracing

- Developed a multi-threaded ray tracing renderer in C++ that distributes pixel computation across a configurable number of threads, boosting image generation by approximately 400% and shows the result on a SFML window.
- Built a JSON-driven configuration system covering four parameter groups — scene objects (size, color, material, position), camera (position, orientation, focal length), image output (dimensions, samples per pixel), and thread count — so scenes can be re-rendered without touching code.

Work Experience

Front End Intern Jan 2024 — Feb 2024
Inbound Agency LTD *Multan, Pakistan*

- Developed UI components using HTML, CSS, JavaScript, and React under senior developer guidance.
- Performed database queries and content updates through a Strapi headless CMS.

Education

Virtual University of Pakistan Oct 2022 — Expected: Dec 2026
Bachelor's of Science, Computer Science *Lahore, Pakistan*

- Cumulative GPA: 3.7/4.0

Achievements

Virtual University of Pakistan Apr 2026
National Skill Competency Test *Multan, Pakistan*

- Achieved Percentile: 90%